A Constructive Introduction to Quantum Field Theory

Lecture 5: Φ^4_3 on the torus
I. The FBSDE in force coordinates, two renormalisations and the large-field problem

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Abstract

The last two lectures construct the massive Φ^4_3 measure on the three-dimensional torus with the forward–backward equations of Lecture 4. In this lecture we set up the problem. For the mass regulator the Polchinski process of Lecture 4 is a noisy observation of the terminal field, and its forecast is the posterior mean; this gives an exact FBSDE at every lattice cutoff. Its backward half is the force equation of Lecture 4 in the forecast coordinate. We cannot solve that equation, so we construct an explicit approximate force and write the exact FBSDE in the coordinates it defines: the *approximate forecast* Y and the *error* Z. The error is driven only by the failure of the approximate force to solve it. We explain why the approximate force needs two renormalisations, in two senses: a linearly and a logarithmically divergent mass counterterm, and two correction steps beyond one loop, which make the defect integrable in the backward equation. Finally we identify the large-field problem: the construction only controls fields of size $r_q = (q/\gamma)^{1/2}$, and must be extended to all fields without losing coercivity. The price is an error that is not small but lives on large fields. Lecture 6 controls it by a priori estimates on the FBSDE and removes the cutoffs.

The results presented in Lectures 4–6 were obtained recently, with the use of AI, in collaboration with Sarah-Jean Meyer.

Table of contents

Introduction . . . 1
10 The problem and the exact FBSDE . . . 2
10.1 Lattice, scales and notation . . . 2
The resolution parameter . . . 3
10.2 The cutoff measure and the goal . . . 3
10.3 The Polchinski process as a filter . . . 4
10.4 The exact force in the forecast coordinate . . . 5
10.5 Approximate forecast and error . . . 5
Reading (10.12) . . . 6
Exercises . . . 6
11 Renormalisation: why two of them . . . 7
11.1 Scales and power counting . . . 7
The basic loop estimate . . . 8
11.2 The seed: one loop . . . 8
11.3 Corrections and the mass . . . 9
Where the iteration comes from . . . 9
The iteration as Picard steps . . . 10
Why the steps approximate the flow . . . 10
The linear part . . . 10
11.4 Two mass renormalisations . . . 12
11.5 Two correction steps . . . 12
11.6 The explicit counterterm . . . 13
Exercises . . . 14
12 The flow at large fields . . . 14
12.1 The large-field problem . . . 14
A scalar model . . . 15
12.2 Clipping . . . 15
12.3 The complete defect . . . 15
Where the large-field error comes from . . . 16
The coordinate correction . . . 16
12.4 Why the Polchinski expansion is not enough . . . 17
The exact force is linear at large fields . . . 17
The field equation does not see the cube . . . 18
What clipping does to the Burgers equation . . . 18
Integrating the cube . . . 19
12.5 The force below the auxiliary scale . . . 21
Exercises . . . 21
References . . . 22

Introduction

Lecture 4 formulated a programme (Figure 7.1 and the remark following it): solve the FBSDE

$$\Phi_t = \int_0^t \dot{Q}_s F_s \, \mathrm{d}s + \int_0^t \dot{Q}_s^{1/2} \mathrm{d}W_s, \qquad F_t = \mathbb{E}_t[-D V_T(\Phi_T)],$$

with a renormalised interaction V_T, with estimates uniform in the ultraviolet cutoff, and take the limit. For Φ^4_3 the force F_t is not expected to converge to an integrable terminal value, and the naive estimates of the force diverge (Appendix A.2 of Lecture 4). In these two lectures we carry out a version of the programme on the unit torus $\mathbb{T}^3$. The guiding principle is the one of Section 8: do not estimate Φ and F directly, but pass to the forecast coordinate, in which the forward equation has no drift and the tree diagrams have been integrated exactly. Section 12.4 shows on one site what goes wrong with a frozen cube, and what integrating it gives.

The plan. Figure 10.1 gives the plan: this lecture builds the approximate force $\bar{\mathcal{F}}$; Lecture 6 does the estimates and the limits. The emphasis is on the structure of the argument. We give complete proofs of the identities and of the scalar models of the estimates, and only describe the tensor estimates on the lattice. The point of view is that of [BCG26]; for other constructions of Φ^4_3 see [GJ73, FO76, BFS83, BG20, GH21].

What is proved. Theorem 10.2 holds for every coupling, every reference mass $q_0 > 0$ and every finite part r of the counterterm, on a fixed torus, with no smallness, and for an explicit counterterm whose finite part is arbitrary. Section 18.7 lists what is not proved. Detailed proofs of what these lectures cite or condense are in the Supplement.

The reference mass. The number $q_0 > 0$ is not an additional regulator: it is the squared mass $m^2 = q_0$ of the Gaussian reference $C = C_{q_0} = (q_0 - \Delta_N)^{-1}$. It must be positive even on the torus, where there is no infrared problem at large distances, because the constants are in the kernel of $-\Delta_N$, so that $(-\Delta_N)^{-1}$ does not exist. The counterterm cannot supply this positivity, since its finite part r is an arbitrary real number, negative values included. The measure ν_N depends on q_0 and μ_N only through $q_0 + \mu_N$, so the two parameters are redundant: one may fix q_0 once and for all, for instance $q_0 = 1$, and let r be the free mass parameter. The number q_0 is also where the scale starts: the Polchinski process runs over $q \in [q_0, \infty)$ and vanishes at $q = q_0$ (see (10.6)).

One auxiliary scale. The expansion of the force needs g_q small, and this happens above an auxiliary scale $q_* \geqslant 1$, depending on λ, q_0 and on the finite part. Below it the coefficients are frozen (Section 12.5), so the force, the system and the results hold at all scales, and no statement about ν depends on it.

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10 The problem and the exact FBSDE

10.1 Lattice, scales and notation

Let N be an odd integer and $\Lambda_N = (N^{-1}\mathbb{Z}/\mathbb{Z})^3$ the periodic lattice of mesh N^{-1} in the unit torus. Let $H_N = \mathbb{R}^{\Lambda_N}$ with

$$\langle u, v \rangle = \int u\, v, \qquad \int f := N^{-3} \sum_{x \in \Lambda_N} f(x),$$

so that the volume is 1 and integrals approximate continuum integrals. The nearest-neighbour Laplacian is $\Delta_N u(x) = N^2 \sum_{i=1}^3 [u(x+e_i/N) + u(x-e_i/N) - 2u(x)]$; we usually drop the index N. A kernel $C(x,z)$ acts by $Cu(x) = \int C(x,z)\,u(z)\mathrm{d}z$, so the identity has kernel $N^3 1_{x=z}$, and

$$\mathrm{Tr}C = \int C(x,x)\mathrm{d}x, \qquad (\mathrm{diag}C)(x) = C(x,x), \qquad \|C\|_{\mathrm{row}} = \sup_x \int |C(x,z)|\mathrm{d}z.$$

The *row norm* $\|C\|_{\mathrm{row}}$ is the largest L^1 norm of a row of the kernel. It bounds the operator norm on L^∞, and for a symmetric kernel on every L^p, $1 \leqslant p \leqslant \infty$, by interpolation: $\|C\|_{p\to p} \leqslant \|C\|_{\mathrm{row}}$. M_f is multiplication by f, and $K + \gamma y^2$ means $K + \gamma M_{y^2}$. For a functional F on H_N, DF, D^2F, D^kF are its gradient and derivatives, as in Lecture 3. $\|u\|_p$ is the L^p norm for the measure $\int$.

The resolution parameter. Lecture 3 used a general covariance profile. Here we use the *mass-regulator profile* of Exercise 8.8 of Lecture 4: with $m^2 = q_0$ and $t \geqslant 0$,

$$\hat{Q}_t(k) = \frac{1}{|k|^2 + q_0} - \frac{1}{|k|^2 + q_0 + t^2}, \qquad \mathcal{R}_t = t^2.$$

It is convenient to parametrise the resolution by the squared regulator mass

$$q := q_0 + t^2 \in [q_0, \infty).$$

Set

$$K_q := q - \Delta_N, \qquad C_q := K_q^{-1}, \qquad C := C_{q_0} = (m^2 - \Delta_N)^{-1}. \tag{10.1}$$

In the notation of Lectures 3–4, with $T = \infty$ and using q instead of t as the resolution variable (so $\dot{Q}$ now means $\partial_q Q_q$),

$$Q_q = C - C_q, \qquad \partial_q Q_q = C_q^2, \qquad \dot{Q}^{1/2} = C_q, \qquad S_q = C_q, \qquad \mathcal{R}_q = q - q_0, \qquad \partial_q \mathcal{R}_q = 1. \tag{10.2}$$

So C_q is the covariance of the scales not yet resolved, and $K_q = S_q^{-1}$ is the precision of the posterior at resolution q under the free prior μ^C. The main properties of C_q are elementary.

Lemma 10.1. (Free resolvents) *For $q \geqslant q_0$ (constants depend on the lower bound only for scales below one), uniformly in N: C_q has a nonnegative kernel, $C_q 1 = q^{-1} 1$, $\|C_q\|_{p\to p} = q^{-1}$ for $1 \leqslant p \leqslant \infty$, and*

$$a_2(q) := (C_q^2)(x,x) = \mathrm{Tr}C_q^2 \lesssim q^{-1/2} \wedge N^3 q^{-2}, \qquad \mathrm{Tr}C_q \lesssim N \wedge N^3 q^{-1}. \tag{10.3}$$

For $y \in H_N$ and $\gamma > 0$, the killed resolvent $G_q(y) = (K_q + \gamma y^2)^{-1}$ satisfies $0 \leqslant G_q(y)(x,z) \leqslant C_q(x,z)$.

Proof. $C_q = \int_0^\infty \mathrm{e}^{-qs} \mathrm{e}^{s\Delta} \mathrm{d}s$ and $\mathrm{e}^{s\Delta}$ is the transition kernel of the lattice random walk. This gives positivity and the row sums. Killing the walk at rate γy^2 gives the comparison for G_q. The symbol of $-\Delta_N$ is comparable to $|k|^2$ on the centred frequency cube $n \in \{-\frac{N-1}{2}, \dots, \frac{N-1}{2}\}^3$ (symbol $4N^2 \sum_i \sin^2(\pi n_i/N)$, with $k = 2\pi n$), and $\sum_k (q + |k|^2)^{-2} \leqslant C \int_{\mathbb{R}^3} (q + |k|^2)^{-2} \mathrm{d}k = C q^{-1/2}$. Counting the N^3 modes gives the other bounds. □

The *bubble* $a_2(q)$ is the coincident-point variance of the shell at resolution q: $\partial_q \mathrm{Var}(X_q(x)) = a_2(q)$ for the free field at resolution q, and $\int_{q_0}^\infty a_2 = \mathrm{Tr}C_{q_0} \asymp N$ is the divergent variance of the lattice field.

10.2 The cutoff measure and the goal

Fix $\lambda > 0$ and put $\gamma := 3\lambda$ (not to be confused with the counterterm γ_T of Lecture 4). The interaction and the lattice measure are

$$U(\varphi) = \frac{\lambda}{4} \int \varphi^4, \qquad V_N(\varphi) = U(\varphi) + \frac{\mu_N}{2} \int \varphi^2, \qquad \nu_N(\mathrm{d}\varphi) = \frac{\mathrm{e}^{-V_N(\varphi)}}{\mathcal{Z}_N}\, \mu^C(\mathrm{d}\varphi), \tag{10.4}$$

where μ^C is the centred Gaussian measure with covariance $C = C_{q_0}$ on H_N, and $\mu_N \in \mathbb{R}$ is a mass counterterm to be chosen. With this normalisation $D^2 U(y) = \gamma M_{y^2}$. In Lecture 4 we wrote the interaction as

$$\lambda \int H_4(\varphi; c_T) + \gamma_T \int H_2(\varphi; c_T);$$

the Wick ordering is now part of μ_N, and the coupling is rescaled by 4. For every real μ_N, ν_N is a probability measure with all moments, since the quartic term dominates.

Theorem 10.2. **(Main result, informal)** *Let $\lambda > 0$, $q_0 > 0$, $r \in \mathbb{R}$, and*

$$\mu_N = r - 3\lambda \operatorname{Tr} C + 6\lambda^2 \int C^3.$$

Then:

a) the laws ν_N, embedded in $H^{-s}(\mathbb{T}^3)$, $s > \frac{1}{2}$, converge along the whole sequence to a measure ν, and ν is an explicit density over the explicit flow from $Y_{q_0} = 0$ (Theorem 18.5);

b) the forecast processes converge in law, along the whole sequence, tested at each scale and jointly with the terminal field, to a solution of a forward–backward system on $[q_0, \infty)$, $Y_{q_0} = 0$, driven by a cylindrical Brownian motion, whose terminal value has the law ν; the forecast is $\psi_q = m_q(\Phi_q)$ with m_q continuous (Theorem 18.6);

c) ν has a renormalised variational formula (Theorem 18.12), and it satisfies the integration by parts formula of Theorem 18.8, in which the renormalised cube appears only tested in expectation, and it has quartic exponential tails for tested fields.

The bounds that prove (a) and (b) are uniform in N: this is *UV stability*. They are bounds on the *error* Z of an approximate forecast, defined below. We now set up the exact FBSDE at fixed N.

10.3 The Polchinski process as a filter

At fixed N everything in Sections 7–8 of Lecture 4 is a theorem. We need it in the profile (10.2), where the regulator is $q - q_0$ and the terminal resolution $q = \infty$.

Proposition 10.3. **(The Polchinski process)** *At every fixed N, the Polchinski process of Theorem 7.3 of Lecture 4 starts at $\Phi_{q_0} = 0$ and solves*

$$\mathrm{d}\Phi_q = C_q^2 F_q \, \mathrm{d}q + C_q \, \mathrm{d}W_q, \qquad F_q = \mathbb{E}_q[-D V_N(\Phi_\infty)], \tag{10.5}$$

where W is a Brownian motion on H_N, $\mathbb{E}_q$ is the conditional expectation given $\mathcal{F}_q = \sigma(\Phi_s : s \leqslant q)$, and $\Phi_q \to \Phi_\infty \sim \nu_N$ as $q \to \infty$ (by (10.6) below; see the proof of Lemma 16.1).

Proposition 10.4. **(Filtering)** *At every fixed N, by Proposition 7.6 of Lecture 4:*

a) the Polchinski process is a noisy observation of the terminal field: there is a Brownian motion B on H_N, independent of Φ_∞, with

$$\Phi_q = (q - q_0) C_q \Phi_\infty + C_q B_{q - q_0}; \tag{10.6}$$

b) the forecast $\psi_q = \mathbb{E}_q \Phi_\infty = \Phi_q + C_q F_q$ is a smooth function of Φ_q, $\psi_q = m_q(\Phi_q)$, with $D m_q = \operatorname{Cov}_q K_q$ invertible, where $\operatorname{Cov}_q = \operatorname{Cov}(\Phi_\infty | \mathcal{F}_q) > 0$;

c) W is the innovation of the filter, and

$$\mathrm{d}\psi_q = \operatorname{Cov}_q \mathrm{d}W_q. \tag{10.7}$$

Proof. In this profile $J_q = (q-q_0)C_q$ and $S_q = C_q$, so (a) is the bridge (7.14) and (c) the filter equation of Lecture 4. For (b), the conditional law of Φ_∞ is ν_N tilted by $\exp(\langle K_q\Phi_q, \varphi\rangle - \frac{q-q_0}{2}\|\varphi\|_2^2)$ by (7.16): an exponential family in $K_q\Phi_q$, whose mean has derivative the covariance. □

The *noise* of the forecast is the conditional expectation of the bridge noise: taking $\mathbb{E}_q$ in (10.6),

$$\eta_q := K_q\Phi_q - (q-q_0)\psi_q = \mathbb{E}_q[B_{q-q_0}],$$

a function of the field, $\eta_q = \eta_q(\Phi_q)$ with $\eta_q(x) = K_q x - (q-q_0)m_q(x)$. It is the noise of the field equation of Lecture 4 in this profile, and it is sub-Gaussian whatever the interaction (Proposition 9.5 of Lecture 4, Lemma 13.2 of Lecture 6).

10.4 The exact force in the forecast coordinate

By Proposition 10.4(b) the map $\Phi_q \mapsto \psi_q$ is invertible, and we define the **exact force in the forecast coordinate** $\mathcal{F}_q$ (not to be confused with the filtration) by

$$\Phi_q = \psi_q + C_q\mathcal{F}_q(\psi_q), \quad \text{i.e.} \quad K_q\Phi_q = K_q\psi_q + \mathcal{F}_q(\psi_q). \tag{10.8}$$

So $F_q = -\mathcal{F}_q(\psi_q)$, the force of Theorem 8.2 of Lecture 4, and the second form is the field equation in the forecast coordinate. We write $\mathrm{Ctr}_E T$ for the contraction of two slots of a tensor T with a kernel E, and $H_q^{\mathrm{ex}}(y) = (K_q + D\mathcal{F}_q(y))^{-1}$.

Proposition 10.5. (The force equation with a mass regulator) *At fixed N, $\mathcal{F}_q$ is smooth, $D\mathcal{F}_q$ is symmetric, $K_q + D\mathcal{F}_q > 0$, and*

$$\partial_q\mathcal{F}_q(y) + \frac{1}{2}\mathrm{Ctr}_{H_q^{\mathrm{ex}}(y)^2} D^2\mathcal{F}_q(y) = 0, \quad \mathrm{Cov}_q = H_q^{\mathrm{ex}}(\psi_q). \tag{10.9}$$

Proof. Differentiating (10.8) in ψ and using $Dm_q = \mathrm{Cov}_q K_q$ gives $K_q + D\mathcal{F}_q(\psi_q) = \mathrm{Cov}_q^{-1}$. Apply Itô's formula to $\Phi_q = \psi_q + C_q\mathcal{F}_q(\psi_q)$ with (10.5), (10.7) and $\mathrm{d}C_q = -C_q^2\,\mathrm{d}q$: the martingale parts agree, and the drifts differ by C_q times the left side of (10.9). At fixed N the forecast has full support, so this holds for all y. □

This is the derivative of the Wetterich equation of Lecture 4 (Exercise 10.1). It is an identity about the unknown ν_N, not a construction. But since $\mathrm{d}\psi_q = H_q^{\mathrm{ex}}(\psi_q)\,\mathrm{d}W_q$, an approximate force gives approximate forecast coordinates, with an error governed by how badly it fails to solve (10.9).

10.5 Approximate forecast and error

Let $\bar{\mathcal{F}}_q$ be, for $q \geqslant q_0$, an odd map on H_N, continuous and piecewise smooth in q and y, with $D\bar{\mathcal{F}}_q$ symmetric and $K_q + D\bar{\mathcal{F}}_q \geqslant \bar{c}_q I$ with $\bar{c}_q > 0$. Define the approximate response, the **force defect** and its drift

$$H_q(y) := (K_q + D\bar{\mathcal{F}}_q(y))^{-1},$$
$$\mathcal{D}_q(y) := \partial_q\bar{\mathcal{F}}_q(y) + \frac{1}{2}\mathrm{Ctr}_{H_q(y)^2} D^2\bar{\mathcal{F}}_q(y), \quad b_q(y) := -H_q(y)\mathcal{D}_q(y). \tag{10.10}$$

The defect is the failure of $\bar{\mathcal{F}}$ to solve (10.9). The map $y \mapsto y + C_q\bar{\mathcal{F}}_q(y)$ is a bijection of H_N. Define the approximate forecast and the error by

$$Y_q + C_q\bar{\mathcal{F}}_q(Y_q) = \Phi_q, \quad Z_q := \psi_q - Y_q, \quad \Xi_q := \mathrm{Cov}_q - H_q(Y_q). \tag{10.11}$$

Equivalently Y_q solves the approximate coordinate equation $K_q\Phi_q = K_q Y_q + \bar{\mathcal{F}}_q(Y_q)$, which is (10.8) with $\bar{\mathcal{F}}$ in place of $\mathcal{F}$. If $\bar{\mathcal{F}} = \mathcal{F}$ then $Y = \psi$, $Z = 0$, $\mathcal{D} = 0$ and $b = 0$.

Theorem 10.6. (The FBSDE in approximate forecast coordinates) *At every fixed N,*

$$\begin{aligned} \mathrm{d}Y_q &= [b_q(Y_q) + H_q(Y_q)\, Z_q]\mathrm{d}q + H_q(Y_q)\, \mathrm{d}W_q, \\ \mathrm{d}Z_q &= -[b_q(Y_q) + H_q(Y_q)\, Z_q]\mathrm{d}q + \Xi_q\, \mathrm{d}W_q, \end{aligned} \tag{10.12}$$

for $q \geqslant q_0$, and $Y + Z = \psi$ is the martingale $\int \mathrm{Cov}\, \mathrm{d}W$.

Proof. Y_q is a smooth function of (q, Φ_q), so it is an Itô process, $\mathrm{d}Y = \beta\, \mathrm{d}q + A\, \mathrm{d}W$. Apply Itô's formula to $Y_q + C_q \bar{\mathcal{F}}_q(Y_q) = \Phi_q$, with (10.5) and $\mathrm{d}C_q = -C_q^2\, \mathrm{d}q$, and multiply by K_q: the martingale parts give $(K + D\bar{\mathcal{F}})A = I$, so $A = H$, and the drifts give

$$(K + D\bar{\mathcal{F}})\beta = CF + C\bar{\mathcal{F}}(Y) - \partial_q \bar{\mathcal{F}} - \frac{1}{2} C \mathrm{tr}_{H^2} D^2 \bar{\mathcal{F}} = \psi - Y - \mathcal{D},$$

since $CF = \psi - \Phi$ and $C\bar{\mathcal{F}}(Y) = \Phi - Y$. Hence $\beta = H(\psi - Y - \mathcal{D}) = HZ + b$, which is the first equation. The second is $\mathrm{d}Z = \mathrm{d}\psi - \mathrm{d}Y$ and (10.7). The computation uses no initial value. At the reference scale $\Phi_{q_0} = 0$, so $Y_{q_0} = 0$ by oddness. □

Reading (10.12). The equation for Y is forward and explicit, once Z is known: its coefficients b, H are deterministic functions built from $\bar{\mathcal{F}}$. The equation for Z is backward. Its terminal condition is not imposed but *proved*: if $\bar{\mathcal{F}}_q$ matches the bare force as $q \to \infty$, then $qZ_q \to 0$ at each fixed N (Lemma 16.1). Its only source is the drift b, i.e. the force defect, transported by the propagator $\exp \int H$. So:

UV stability means that the explicit force $\bar{\mathcal{F}}$ computes the forecast up to an error Z which is small uniformly in N.

For this we need: (i) a force defect $\mathcal{D}$ integrable against the backward propagator, which is a deterministic renormalisation problem (Section 11); (ii) global coercivity of $\bar{\mathcal{F}}$, so that $H_q(y)$ is a good noise coefficient for all fields (Section 12); (iii) a priori estimates for the coupled system (Lecture 6). Figure 10.1 summarises the logic of the two lectures.

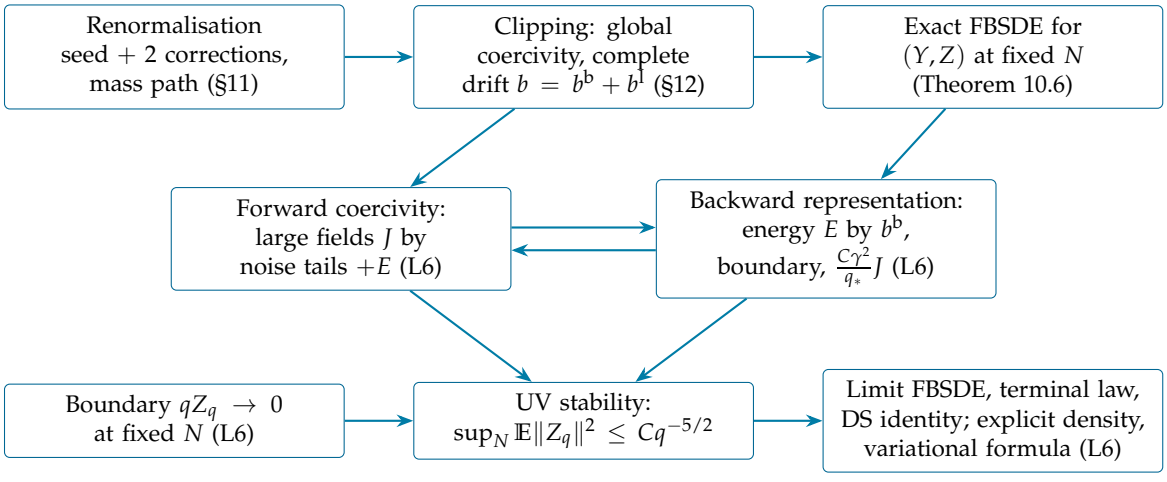


Figure 10.1. The structure of the construction (E, J: the energies of Lecture 6). The two arrows between the forward and backward estimates form the closed loop of Lecture 6: neither estimate closes on its own.

Exercises

Exercise 10.1. (Same force) Show that $\mathcal{F}_q$ of (10.8) coincides with the $\mathcal{F}_t$ of (8.5) of Lecture 4 for the profile (10.2). Show that $\mathcal{F}_q = D\Gamma_q$, where $\partial_q \Gamma_q = \frac{1}{2} \mathrm{Tr}\, H_q^{\mathrm{ex}}$, and that this agrees with the Wetterich equation (8.13), the difference being $-\frac{1}{2} \mathrm{Tr}\, C_q$.

Exercise 10.2. (**Gaussian case**) Let $V_N(\varphi)=\frac{a}{2}\|\varphi\|^2$, $a>0$. Compute ψ_q, Cov_q and $\mathcal{F}_q$ explicitly, and check Proposition 10.4 and (10.9). Compare with Exercise 7.1.

Exercise 10.3. (**Bayes for the whole path**) Show that the conditional law of Φ_∞ given the whole path $(\Phi_s)_{s\leqslant q}$ depends only on Φ_q: the Polchinski process is a sufficient statistic for itself (Proposition 7.6(a) of Lecture 4).

Exercise 10.4. (**One site**) Take $N=1$ (one site, $\Delta=0$, $K_q=q$) and $\bar{\mathcal{F}}_q(y)=\lambda y^3$. (a) Show that Y_q is the real root of $qY+\lambda Y^3=q\Phi_q$, and that $H_q(y)=(q+\gamma y^2)^{-1}$. Explain how large fields suppress the noise of Y. (b) Compute the defect $\mathcal{D}_q(y)=\gamma y(q+\gamma y^2)^{-2}$ and the drift $b_q(y)=-\gamma y(q+\gamma y^2)^{-3}$. Show that $|b_q(y)|\leqslant\gamma q^{-3}|y|$, and compare with Proposition 12.2: on one site $a_2(q)=q^{-2}$ instead of $q^{-1/2}$, and no renormalisation is needed.

11 Renormalisation: why two of them

We now construct the force $\bar{\mathcal{F}}_q$ on bounded fields. This section is deterministic: it uses no information on ν_N.

11.1 Scales and power counting

Two scales govern everything:

$$g_q:=\gamma q^{-1/2},\qquad r_q:=(q/\gamma)^{1/2},\qquad \gamma r_q^2=q. \tag{11.1}$$

g_q is the dimensionless coupling at resolution q: in $d=3$ the coupling λ has dimension of mass, and $q^{1/2}$ is the mass of the regulator. It is small at all scales $q\geqslant q_*$ if $g_{q_*}=\gamma q_*^{-1/2}$ is small; this is the only large number in the construction, and it is not a physical mass. r_q is the field size at which the nonlinearity is comparable to the regulator: $D^2U(y)=\gamma y^2\approx q$ when $|y|\approx r_q$. Compare with the size of the free field at resolution q:

$$\mathrm{Var}X_q(x)=\int_{q_0}^{q}a_2\leqslant Cq^{1/2},$$

so typical fields have size $q^{1/4}=g_q^{1/2}r_q\ll r_q$.

We measure a k-tensor $T(x_1,\dots,x_k)$ (for instance a **vertex** with k legs, the kernel of the $(k-1)$th field derivative of a force) by its rooted norm

$$s(T)=\max_i\sup_{x_i}\int|T(x_1,\dots,x_k)|\prod_{j\neq i}\mathrm{d}x_j,$$

and a force g by

$$\mathsf{S}_k(g)=\sup_{y\in\mathcal{B}_q}s(D^{k-1}g(y))$$

on the field ball

$$\mathcal{B}_q:=\big\{y\in H_N:\|y\|_\infty<2r_q\big\}.$$

The natural size of the k-leg vertex of the loop corrections

$$\bar{\mathcal{F}}_q-U'-M_qy$$

to the bare force ($U'=DU=\lambda y^3$ the bare force, M_q a linear mass term, Section 11.3) is

$$\mathsf{S}_k\lesssim g_q\,q\,r_q^{2-k}. \tag{11.2}$$

For $k=2$ this says the derivative of the correction force is g_q times the regulator q (while $D^2U\approx q$ at $|y|\approx r_q$); each extra field derivative costs r_q^{-1}. For $k=4$, $g_q\,q\,r_q^{-2}=\gamma g_q$: the four-leg vertex of the corrections is γ times the small coupling. The whole construction is a bookkeeping of (11.2).

The basic loop estimate. The loop $\frac{1}{2}\mathrm{Ctr}_{H^2} D^2\bar{\mathcal{F}}$ and its derivatives are closed chains

$$H\,(D^{j_1+1}\bar{\mathcal{F}})H\cdots(D^{j_p+1}\bar{\mathcal{F}})H.$$

A closed loop with p vertices, each carrying at least one external leg (vertices without external legs are resummed into H), and $p+1$ propagators H (two of them adjacent, H^2, once the loop is closed) is estimated by rooting one external slot, using two propagators in L^2 (each row has L^2 norm $\leqslant (a_2)^{1/2}$) and the others in operator norm ($\leqslant C/q$):

$$s(\text{loop}) \leqslant C\, a_2\, q^{-(p-1)} \prod_{i=1}^{p} s(T_i) \tag{11.3}$$

Indeed, by cyclicity of the trace we may take the rooted slot x on T_1. Integrating out the other external legs, each T_i, $i \geqslant 2$, is dominated by a kernel $t_i(u,v)$ in its two internal legs whose rows and columns have L^1 norm $\leqslant s(T_i)$, so $\|t_i\|_{2\to 2} \leqslant s(T_i)$ by Schur's test; T_1 with x fixed gives a kernel t_1^x with $\int t_1^x \leqslant s(T_1)$. The rest of the loop is a kernel $|H|R|H|$, where the two propagators next to T_1 are kept apart and R collects the other $p-1$ propagators and $t_2,\dots,t_p$. By Cauchy–Schwarz its entries are bounded by

$$\sup_x \|H(x,\cdot)\|_2^2 \|R\|_{2\to 2} \leqslant C a_2\,(C/q)^{p-1} \prod_{i\geqslant 2} s(T_i),$$

and integrating against t_1^x gives (11.3). We used only that the propagators are symmetric with rows of L^2 norm $\leqslant C a_2^{1/2}$ and of L^1 norm $\leqslant C/q$, which also bounds their operator norm on L^2. For the free resolvent these are $\int C_q(x,z)^2 \mathrm{d}z = a_2$ and $\int C_q(x,z)\mathrm{d}z = q^{-1}$ (Lemma 10.1); for the propagators used below they are (12.3), and Lemma 20.1 of the Supplement.

The vertices of the bare force have

$$s\,(D^{j+2}U) \leqslant C\, q\, r_q^{-j}$$

on the ball ($D^3U = 2\gamma y$, $D^4U = 2\gamma$). With $a_2 \leqslant C q^{-1/2}$, a loop with k external legs built from them is $\leqslant C q^{1/2} r_q^{-k}$, and after the integration in q that produces the force $\bar{\mathcal{F}}$ (which gains a factor q) it has size $q^{3/2} r_q^{-k} = g_q q r_q^{2-k}$: this is the origin of (11.2), the size of one loop. Replacing a bare vertex by a vertex of a correction of relative size ε multiplies the loop by ε (Exercise 11.1). *Every further loop gains one power of g_q.*

11.2 The seed: one loop

The first approximation is the one-loop formula of Exercise 8.6(b): replace $D\bar{\mathcal{F}}$ by $DU' = \gamma y^2$ in the force equation and integrate from $q = \infty$. With the linear part subtracted this gives the seed

$$\ell_q(y)(x) = \gamma y(x)[G_q(y)(x,x) - C_q(x,x)], \quad \partial_q \ell_q(y)(x) = -\gamma y(x)[G_q(y)^2(x,x) - a_2(q)], \tag{11.4}$$

with $G_q(y) = (K_q + \gamma y^2)^{-1}$. It is odd, and its linear Taylor coefficient at $y = 0$ vanishes. Its expansion in powers of the field,

$$\ell_q(y)(x) = \gamma y(x) \sum_{j\geqslant 2} (-1)^{j-1} [C_q(\gamma y^2 C_q)^{j-1}](x,x),$$

is the sum of one-loop diagrams with $2j$ external legs, one of them the output x. The leading term is the bubble

$$-\gamma^2 y(x) \int C_q(x,z)^2 y(z)^2\,\mathrm{d}z,$$

with

$$s(D^3\ell_q(0)) = 6\gamma^2 a_2(q) \leqslant C g_q q r_q^{-2},$$

as predicted by (11.2).

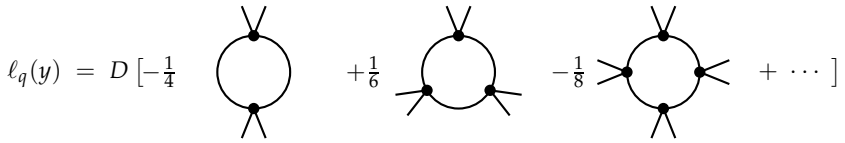


Figure 11.1. The seed (11.4) as one-loop graphs: lines C_q, and a vertex γy^2 with its two external fields at each corner. The j-th ring has $2j$ legs and coefficient $(-1)^{j+1}/(2j)$. Marking one leg of a ring gives its term in the seed force. The series sums to a logarithm and has radius of convergence r_q. The quartic ring is the bubble.

11.3 Corrections and the mass

The seed solves the force equation only with DU' inside the loop. Its defect is of relative size g_q compared with the one-loop flow $q^{1/2}r_q^{-k}$ (two loops). To improve it, let

$$\Pi g(y) = g(y) - Dg(0)y$$

remove the linear Taylor coefficient (the low jet) of an odd force, let

$$M_q = \mu_q I + \Theta_q$$

be a linear operator path (a running mass μ_q and an even, translation-invariant, centred kernel Θ_q, $\Theta_q 1 = 0$), and set

$$\mathcal{J}_q[M,\ell](y) := -\frac{1}{2}\mathrm{Ctr}_{R^2} D^2(U'+\ell)(y), \quad R = (K_q + M_q + D(U'+\ell)(y))^{-1}. \tag{11.5}$$

The approximate force on the field ball is

$$\mathcal{F}_q^o(y) = U'(y) + M_q y + \ell_q^{[2]}(y), \tag{11.6}$$

with the nonlinear part built by two integrated steps of the force equation,

$$\ell^{[0]} = \ell, \quad \ell_q^{[j+1]} = \ell_q - \int_q^\infty \Pi(\mathcal{J}_s[M,\ell^{[j]}] - \partial_s \ell_s)\,\mathrm{d}s, \quad j = 0,1, \tag{11.7}$$

so that

$$\partial_q \ell^{[j+1]} = \Pi \mathcal{J}_q[M,\ell^{[j]}]:$$

each step inserts the previous approximation into the loop. The integrals run from ∞: the nonlinear part is fixed by the requirement that it vanishes at infinite resolution, where $\mathcal{F}_q$ must become the bare force (Theorem 8.2).

Where the iteration comes from. The exact force solves the force equation (10.9), $\partial_q \mathcal{F}_q = -\frac{1}{2}\mathrm{Ctr}_{H_q^{\mathrm{ex}2}} D^2 \mathcal{F}_q$, and becomes the bare force $DV_N = U' + \mu_N y$ at $q = \infty$. Write an odd candidate as $\mathcal{F}_q = U' + M_q y + \ell_q$: the bare cube, a linear part, and a nonlinear remainder without linear Taylor coefficient, $\Pi\ell_q = \ell_q$. Since $\partial_q U' = 0$, $D^2(M_q y) = 0$ and $D\mathcal{F}_q = M_q + D(U' + \ell_q)$, the force equation reads, with the trace map (11.5),

$$(\partial_q M_q)y + \partial_q \ell_q = \mathcal{J}_q[M,\ell_q](y).$$

The projections Π and $1-\Pi$ split it into two equations of different nature:

$$\partial_q \ell_q = \Pi \mathcal{J}_q[M,\ell_q], \qquad (\partial_q M_q)y = D\mathcal{J}_q[M,\ell_q](0)y.$$

The first is the nonlinear part of the flow. By power counting (11.2) its vertices with $k \geqslant 4$ legs have integrands $O(s^{-k/2})$, integrable at $s=\infty$, so it can be integrated from infinite resolution with the boundary value $\ell_\infty = 0$ required by the bare force:

$$\ell_q = -\int_q^\infty \Pi \mathcal{J}_s[M, \ell_s]\, \mathrm{d}s.$$

This is a fixed-point equation: ℓ appears on both sides, inside the propagator R and in $D^2\ell$. The second equation is the linear part of the flow. It is not integrable at $q=\infty$ uniformly in N, since the tadpole and the sunset diverge (Section 11.4). It is solved exactly, as the mass equation (11.8) with a boundary value at q_*, and its value at $q=\infty$ is the bare mass. This is the reason for the subtraction Π: it separates the convergent nonlinear flow, which we approximate, from the divergent linear flow, which we solve exactly and which produces the counterterm.

The iteration as Picard steps. The recursion (11.7) is the Picard iteration of this fixed-point equation. The seed is its first step: with $\ell=0$ and $M=0$ inside the loop, $\mathcal{J}_s[0,0] = -\frac{1}{2}\mathrm{Ctr}_{G_s^2} D^2 U'$, and $\Pi\mathcal{J}_s[0,0] = \partial_s \ell_s$ is (11.4). Since the seed vanishes at infinite resolution, $\ell_q = -\int_q^\infty \partial_s \ell_s\, \mathrm{d}s$, and (11.7) is the same as $\ell_q^{[j+1]} = -\int_q^\infty \Pi \mathcal{J}_s[M, \ell^{[j]}]\, \mathrm{d}s$. It is written as a correction of the seed because its integrand, the difference of two loops, is one order smaller.

Why the steps approximate the flow. Each step gains one loop and one power of g_q. The loop map $\ell \mapsto \mathcal{J}[M,\ell]$ is Lipschitz with a small constant: by the loop estimate (11.3), a change of ℓ of size ε (in units of the natural size $q r_q^{2-k}$ of the k-leg vertex) changes the integrated loop by εg_q in the same units (part (a) of the proof below). So the increments decrease geometrically, as in (11.11), and the iterates approach the fixed point. The quality of $\mathcal{F}^o = U' + My + \ell^{[2]}$ is measured by its defect, the failure to solve the force equation. Since $\partial_q \ell^{[2]} = \Pi \mathcal{J}_q[M, \ell^{[1]}]$ and the mass equation solves the linear part exactly,

$$\partial_q \mathcal{F}_q^o - \mathcal{J}_q[M, \ell^{[2]}] = \Pi(\mathcal{J}_q[M, \ell^{[1]}] - \mathcal{J}_q[M, \ell^{[2]}]),$$

the loop difference between the last two steps, of order g_q^3 by (11.12). This is what the FBSDE sees: by Theorem 10.6 the error Z is driven only by the defect, and Section 11.5 shows that two steps are the minimum for a source that is integrable in the backward equation. The iteration cannot be continued indefinitely: each step consumes two derivatives (see the end of this section), so the number of steps is fixed in advance.

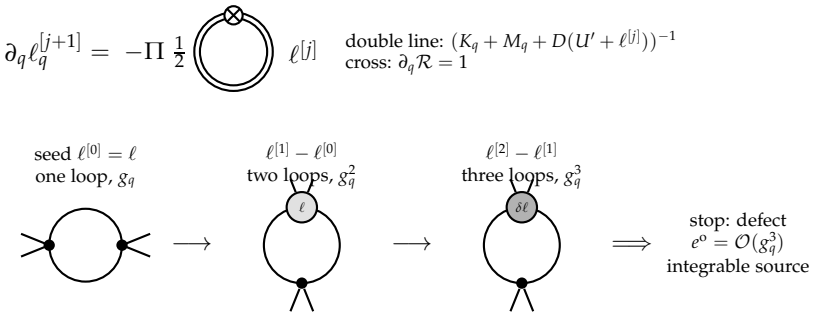


Figure 11.2. The iteration (11.7), schematically, with only two external legs drawn per vertex. Each step inserts the previous correction into the loop and gains one loop and one factor g_q, as in (11.11). After two steps the defect is the loop difference of order g_q^3, which gives (11.12) and makes the source integrable in the backward equation.

The linear part. The low jets are not iterated: they are handled exactly by the path M. The linear part of the flow at $y=0$ is, by oddness,

$$D\mathcal{J}_q(0) = -\frac{1}{2}\mathrm{Ctr}_{R_0^2} D^3 \mathcal{F}^o(0),$$

where $R_{0,q}=(K_q+M_q)^{-1}$. Requiring that M solves this part of the flow exactly gives the mass equation

$$\partial_q M_q = B_q[M] := -\gamma \mathrm{Tr}(R_{0,q}^2)\, I - \frac{1}{2}\mathrm{Ctr}_{R_{0,q}^2} D^3 \ell_q^{[2]}(0), \tag{11.8}$$

(the source $B_q[M]$ is not the sampling Brownian motion B), with the *mixed boundary conditions*

$$\mu_{q_*} = b, \qquad \Theta_q \to 0 \text{ as } q \to \infty$$

(the latter in first spatial moment). The running mass is pinned at the auxiliary scale q_*, at a value b with $|b| \leqslant B\gamma q_*^{1/2}$, while the nonlocal kernel is pinned at infinity. The bare mass is then

$$\mu_N = \lim_{q\to\infty} \mu_q = b + \int_{q_*}^{\infty} \beta_q \, \mathrm{d}q, \qquad B_q 1 = \beta_q 1, \tag{11.9}$$

and it is this number that enters the Gibbs measure (10.4).

Note the order: the counterterm is an output of the deterministic flow, not an input.

Proposition 11.1. (The coefficient construction) *For q_* large enough (depending on γ, not on N), the system (11.7)–(11.8) has a unique solution in a ball, with*

$$|\mu_q| \leqslant C\gamma\sqrt{q}, \qquad v_1(\Theta_q) \leqslant C\gamma^2 q^{-1/2}, \qquad \|\partial_q M_q\|_{\mathrm{row}} \leqslant C g_q, \tag{11.10}$$

where $v_1(\Theta) = \int d(0,z)\,|\Theta(0,z)|\mathrm{d}z$. The seed obeys $\mathcal{N}_{k,q}(\ell) \leqslant C_k g_q q r_q^{2-k}$, $4 \leqslant k \leqslant 10$, and the corrections obey

$$\mathcal{N}_{k,q}(\ell^{[j]} - \ell^{[j-1]}) \leqslant C_k\, g_q^{j+1}\, q\, r_q^{2-k}, \qquad 4 \leqslant k \leqslant 10-2j, \quad j=1,2, \tag{11.11}$$

where $\mathcal{N}_{k,q}(f) = \mathsf{S}_k(f) + \sqrt{q}\; v_1(D^{k-1} f(0))$, and for a k-tensor

$$v_1(T) = \max_{i,a<b} \sup_{x_i} \int d(x_a,x_b)|T| \prod_{j\neq i} \mathrm{d}x_j.$$

The complete defect of $\mathcal{F}^{\mathrm{o}}$ on the ball is

$$\mathcal{D}_q^{\mathrm{o}} = \Pi\,(\mathcal{J}_q[M,\ell^{[1]}] - \mathcal{J}_q[M,\ell^{[2]}]),$$

and

$$\mathsf{S}_4(\mathcal{D}_q^{\mathrm{o}}) \leqslant C\gamma^5 q^{-3}. \tag{11.12}$$

At fixed N the bare mass exists, $|\mu_N - b| \lesssim \gamma \mathrm{Tr} C_{q_*} \lesssim \gamma N$, and $|\mu_N - \mu_q| \leqslant C\gamma N^3/q$, $\|\Theta_q\|_{\mathrm{row}} \leqslant CN\gamma^2 q^{-1/2} \to 0$ as $q \to \infty$.

Idea of the proof (complete proof: Proposition 20.6 of the Supplement). (a) *Loop Lipschitz estimate.* If two corrections $\ell, \hat{\ell}$ obey the individual bounds (11.2) up to order $k+2$ and their difference obeys them with g_q replaced by ε_q, then $\mathcal{N}_k(\mathcal{J}[M,\ell] - \mathcal{J}[M,\hat{\ell}]) \leqslant C\varepsilon_q q^{1/2} r_q^{-k}$. Expand D^{k-1} of the loop force into loops; in each, telescope the difference onto one marked vertex or propagator and apply (11.3). (b) *Integration.* Each step gains one g; the integrand of (11.7) is $O(\gamma^{1+k/2} s^{-k/2})$ in the k-leg vertex, and that of the increment $\ell^{[j+1]} - \ell^{[j]}$ is $O(\gamma^{1+k/2} s^{-k/2} g_s^j)$, integrable at $s=\infty$ for $k \geqslant 4$. The low jets have been removed, so the lower derivatives are controlled by Taylor's formula from the four-leg vertex. (c) *Mass path.* Given M in the ball, the right side of (11.8) is computed from the corrections, and the map $M \mapsto (\mu, \Theta)$ is a contraction for large q_*: differences gain a factor g_{q_*} (for μ after integrating from q_*, using $\sup_x \log x/\sqrt{x} < \infty$). (d) *Defect.* The mass equation supplies exactly the linear jet of the flow, and $\partial_q \ell^{[2]} = \Pi\,\mathcal{J}[M,\ell^{[1]}]$, so the defect is the loop difference between the last two approximations, of order g^3: by (a) with $\varepsilon_q = g_q^3$, $\mathsf{S}_4(\mathcal{D}^{\mathrm{o}}) \leqslant C g_q^3 q^{1/2} r_q^{-4} = C\gamma^5 q^{-3}$. □

Two technical points are hidden in this sketch. First, the loop estimate (a) needs more derivatives than it controls: the k-leg vertex of the loop involves vertices with $k+2$ legs. Ten legs of the seed leave eight after one step and six after two; this is why the number of steps must be finite and fixed in advance (an infinite iteration loses all derivatives). Second, the nonlocal kernel Θ_q must be measured in a first spatial moment v_1, not in a second one: the estimate $\|\delta_z^2 C_q\|_{\mathrm{row}} \lesssim d(0,z)^2$, which a second moment would require, fails by a logarithm, and one uses $\|\delta_z^2 C_q\|_{\mathrm{row}} \lesssim q^{-1/2} d(0,z)$ instead (Lemma 19.3 of the Supplement). This is also why $\mathcal{N}_k$ contains a spatial moment at $y=0$.

11.4 Two mass renormalisations

The bare mass μ_N diverges as $N \to \infty$. Its two divergent parts are the two divergent self-energy diagrams of Appendix A.2 of Lecture 4.

- *The tadpole.* The first term of (11.8) integrates to $-\gamma \int_{q_*}^{\infty} \mathrm{Tr} R_{0,q}^2 \, \mathrm{d}q \approx -\gamma \mathrm{Tr} C = -3\lambda \mathrm{Tr} C \sim -N$ (up to a term bounded uniformly in N). This is Wick ordering: $\frac{\lambda}{4}\varphi^4 - \frac{3}{2}\lambda\, c\, \varphi^2$ agrees, up to a constant, with $\frac{\lambda}{4} H_4(\varphi; c)$, with $c = C(x,x)$ ((A.5) of Lecture 4).
- *The sunset.* The second term contracts the bubble contained in $D^3\ell^{[2]}(0)$ (the cubic coefficient of the seed force) with R_0^2, and integrating in q produces the two-loop self-energy, which diverges like $\lambda^2 \int_{|z|>1/N} |z|^{-3} \mathrm{d}z \sim \lambda^2 \log N$. This is the sunset counterterm $\gamma_T \sim \log T$ of (A.7) of Lecture 4: with $\lambda_{\mathrm{L4}} = \lambda/4$ and $\gamma_T \int H_2 = \gamma_T \int \varphi^2 + \mathrm{const}$, the sunset part of μ_N is $2\gamma_T = 6\lambda^2 \int C^3 \approx \frac{3\lambda^2}{8\pi^2} \log N$, positive. (The double-tadpole logarithms in the two terms of (11.8), the bubble with a tadpole insertion and the mass insertion in $\mathrm{Tr} R_{0,q}^2$, cancel each other.)

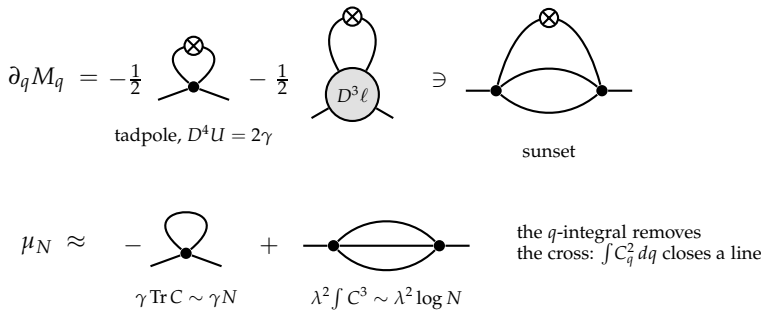


Figure 11.3. The mass equation (11.8) and the two divergent parts of the bare mass. The cross is the insertion $R_{0,q}^2$. Since $D^3\ell^{[2]}(0)$ contains the bubble of Figure 11.1, its contraction contains the sunset. Integrating in q turns the crossed line into a line C: the tadpole is linear in N and the sunset logarithmic.

Power counting (Exercise A.3 of Lecture 4) says that no other counterterm is needed. In the flow this is the statement that the integrals (11.7) of all vertices with $k \geqslant 4$ legs converge at $q = \infty$: only the linear jet must be fixed by a boundary condition.

There is a subtlety: the running mass μ_q at fixed q has a limit as $N \to \infty$, but $\mu_N = \mu_\infty$ does not. One never interchanges these two limits. Also, the *whole* running kernel Θ_q is kept: replacing the sunset by its local part would leave a nonlocal remainder of the wrong size.

11.5 Two correction steps

Why two corrections, and not one or three? The answer is in the backward equation of (10.12). Freeze the noise and the large fields, and estimate the size of Z produced by a defect on the ball. The propagator of $\mathrm{d}Z = -H Z \, \mathrm{d}q + \dots$ from q to s is about s/q (since $H \approx C_q$ has norm q^{-1}), so

$$Z_q \approx \int_q^{\infty} \frac{s}{q}\, b_s \, \mathrm{d}s, \qquad |b_s| \lesssim \|H_s\|\, \mathsf{S}_4(\mathcal{D}_s)\, r_s^3 \lesssim s^{-1} \mathsf{S}_4(\mathcal{D}_s)\, r_s^3, \tag{11.13}$$

where we used that the low jet of $\mathcal{D}$ vanishes, so on the ball $|\mathcal{D}| \leqslant \mathsf{S}_4(\mathcal{D})|y|^3/6$. After n correction steps the defect is the loop difference of the last two approximations, of relative size g^{n+1}: by the loop estimate,

$$\mathsf{S}_4(\mathcal{D}_s) \lesssim g_s^{n+1} s^{1/2} r_s^{-4} = \gamma^{n+3} s^{-(n+4)/2}, \qquad |b_s| \lesssim \gamma^{n+3/2} s^{-(n+3)/2},$$

and therefore

$$|Z_q| \lesssim \frac{\gamma^{n+3/2}}{q} \int_q^\infty s^{-(n+1)/2} \mathrm{d}s.$$

steps n	$\mathsf{S}_4(\mathcal{D}_s)$	$\lvert b_s\rvert$	$\int_q^\infty \frac{s}{q}\lvert b_s\rvert \mathrm{d}s$
0 (seed only)	$\gamma^3 s^{-2}$	$\gamma^{3/2} s^{-3/2}$	∞ (power)
1	$\gamma^4 s^{-5/2}$	$\gamma^{5/2} s^{-2}$	∞ (logarithm)
2	$\gamma^5 s^{-3}$	$\gamma^{7/2} s^{-5/2}$	$2\gamma^{7/2} q^{-3/2}$

With one step the bounded-field source produces a logarithmically divergent error; two steps are the minimum, and they suffice: $Z_q^{\mathrm{b}} = O(q^{-3/2})$, and the backward energy $\gamma^{-1}\int q^{1/2}|Z^{\mathrm{b}}|^2 \mathrm{d}q$ is finite (Lecture 6). The general (fractional) theory gives the depth criterion $(n+2)\,\varkappa > (3+\varkappa)/2$, with subcriticality $\varkappa = \frac{1}{2}$ for Φ_3^4: $(2+2)\cdot\frac{1}{2} = 2 > \frac{7}{4}$, while $n=1$ gives $\frac{3}{2} < \frac{7}{4}$. This is a sufficient condition, which loses $\varkappa/2$; for Φ_3^4 the heuristic above shows that $n>1$ suffices, with $n=1$ borderline.

The two senses of "two renormalisations" are related. The mass counterterm must contain the sunset because the quartic vertex of the corrections, already of the seed, contains the bubble, and contracting it produces the sunset; and the second step is needed because the defect of the first one, the three-loop remainder, is borderline in the backward equation, exactly as the sunset is borderline in the mass.

11.6 The explicit counterterm

Formula (11.9) gives μ_N as an output of the flow, from the boundary value b. The input can be made explicit, and this also fixes what the finite part is.

Theorem 11.2. (The explicit counterterm) *Let $a_2 = \operatorname{Tr} C_q^2$ and $C_q^{(2)}$ be the kernel of C_q^2; in integrals, powers of kernels are pointwise, $\int C_q^2 C_q^{(2)} = \int C_q(0,z)^2 C_q^{(2)}(0,z)\mathrm{d}z$. For $|b| \leqslant B\gamma q_*^{1/2}$ and q_* large,*

$$\beta_q = -\gamma a_2 - \gamma\mu_q a_2' + \gamma^2 a_2^2 + 2\gamma^2 \int C_q^2 C_q^{(2)} + \mathcal{R}_q, \quad |\mathcal{R}_q| \leqslant C\gamma^3 q^{-3/2}, \tag{11.14}$$

and so, for $f_N(b) = \mu_N(b) + 3\lambda \operatorname{Tr} C - 6\lambda^2 \int C^3$,

$$\begin{aligned} f_N(b) =& b(1+\gamma a_2(q_*)) + \gamma(\operatorname{Tr} C - \operatorname{Tr} C_{q_*}) \\ & -6\lambda^2 \int (C^3 - C_{q_*}^3) + \int_{q_*}^\infty (\mathcal{R}_q + \gamma(\beta_q + \gamma a_2) a_2)\, \mathrm{d}q. \end{aligned} \tag{11.15}$$

Moreover $|f_N(b_1) - f_N(b_2) - (b_1 - b_2)| \leqslant C\, g_{q_*} |b_1 - b_2|$, and $f_N \to f$ pointwise along the whole sequence. Consequently, for B above a universal constant, every $r \in \mathbb{R}$ and q_* large depending on λ, q_0, B, r, there is a unique b_N with

$$\mu_N = r - 3\lambda \operatorname{Tr} C + 6\lambda^2 \int C^3, \tag{11.16}$$

and b_N converges along the whole sequence.

Proof. Expansion. Expand $R_0 = C - CMC + CMCMR_0$ with $C = C_q$. Since $a_2' = -2\mathrm{Tr}\, C^3$, the first term of (11.8) is $-\gamma a_2 - \gamma\mu a_2' + 2\gamma \mathrm{Tr}(C^3\Theta)$, up to $O(\gamma a_2 g_q^2)$, and $|\mathrm{Tr}(C^3\Theta)| \leqslant C q^{-1} v_1(\Theta)$ because $|\hat\Theta(k)| \leqslant 2|k| v_1$. The corrections in $\ell^{[2]}$ give $O(\gamma^3 q^{-3/2})$ after contraction, by (11.11). The cubic part of the seed is $-\gamma^2 y(x) \int C^2(x,z) y(z)^2$, and contracting its third derivative at zero with R_0^2 gives the operator $\gamma^2 a_2 \mathrm{diag}(R_0^2) + 2\gamma^2 C^2 \cdot R_0^2$, where $\cdot$ is the pointwise product of kernels. Its constant-mode eigenvalue, with R_0^2 replaced by C^2 at a relative cost g_q, gives the remaining two terms.

Integration. Use $\partial_q \mathrm{Tr}\, C_q = -a_2$, $\partial_q \int C_q^3 = -3 \int C_q^2 C_q^{(2)}$, and $-\gamma\mu a_2' + \gamma^2 a_2^2 = -\gamma \partial_q(\mu a_2) + \gamma(\beta + \gamma a_2) a_2$. This is the cancellation of the double-tadpole logarithms: $\beta + \gamma a_2 = O(\gamma^2/q)$. At fixed N the boundary terms at $q = \infty$ vanish, which gives (11.15).

Response. For two boundary values with difference δb, put $D = \sup_q |\delta\mu_q|$ and $X = \sup_q q^{3/2} v_1(\delta\Theta_q)$, which are finite at fixed N. The loop comparison gives $|\delta\beta_q| \leqslant C\gamma(D q^{-3/2} + X q^{-5/2})$ and $v_1(\delta B_q - \delta\beta_q I) \leqslant C\gamma^2(D q^{-5/2} + X q^{-7/2})$, so $X \leqslant C\gamma^2 D + C\gamma^2 q_*^{-1} X$ and $D \leqslant |\delta b| + C\gamma q_*^{-1/2} D + C\gamma q_*^{-3/2} X$. Hence $D \leqslant 2|\delta b|$, and integrating $\delta\beta$ gives the claim.

Limit and inversion. In (11.15), $\mathrm{Tr}\, C - \mathrm{Tr}\, C_{q_*}$ has summable terms $\lesssim (1+|k|)^{-4}$; $C - C_{q_*}$ converges uniformly and $C^2 + C C_{q_*} + C_{q_*}^2$ in L^1; and the integrand converges at fixed q by consistency of the coefficients, dominated by $C\gamma^3 q^{-3/2}$. As $\mathrm{Tr}\, C - \mathrm{Tr}\, C_{q_*} \leqslant C + q_*^{1/2}/10$ uniformly in N, the range of f_N contains r for B above a universal constant and q_* large. Monotonicity and pointwise convergence give the convergence of b_N. □

The boundary value b is not the finite part: by (11.15), $f_N(b) \approx b + \gamma q_*^{1/2}/(4\pi)$ at leading order, so at fixed b, raising q_* changes the theory.

Exercises

Exercise 11.1. (**Counting loops**) (a) Use (11.3) to show that a loop with p vertices $D^{j_i+1}(U' + \ell)$, $j_i \geqslant 1$, each with $s \leqslant C q r_q^{-j_i}$, and $k = \sum j_i$ external legs has rooted norm $\leqslant C a_2(q) q r_q^{-k} \leqslant C q^{1/2} r_q^{-k}$. (b) Show that $q \cdot q^{1/2} r_q^{-k} = g_q q r_q^{2-k}$. (c) If one vertex is replaced by a vertex of size $\varepsilon q r_q^{-j}$, show that the loop gains the factor ε. Deduce the sizes g^{j+1} of the increments in (11.11).

Exercise 11.2. (**The bubble**) Compute $D^3 \ell_q(0)$. Show that its rooted norm is $6\gamma^2 a_2(q)$ and its first spatial moment is $O(\gamma^2 q^{-1})$.

Exercise 11.3. (**Small derivatives are not enough**) On one site take $q = \gamma = 1$, $M = 0$, fixed small $\alpha, \delta > 0$, $\hat\ell_n(y) = \alpha n^{-1}(n y - \sin(n y))$, and $\ell_n = \hat\ell_n + \frac{\delta}{3} y^3$. Show that both have small derivatives on $|y| < 2$, that $\ell_n - \hat\ell_n$ has bounded derivatives, but $D^3(\mathcal{T}[\ell_n] - \mathcal{T}[\hat\ell_n])(0) = 12(2\delta + \alpha\delta n^2 + \delta^2)$ is unbounded in n. Conclude that the loop estimate needs the individual higher derivative bounds.

Exercise 11.4. (**Why two steps**) Verify the table above. With one step, show that the bounded-field part of Z is of size $q^{-1} \log(T/q)$ on $[q, T]$, for an upper scale cutoff T.

12 The flow at large fields

12.1 The large-field problem

All the estimates of Proposition 11.1 hold on the field ball $\|y\|_\infty < 2 r_q$, and only there. But the forecast Y_q is a random field on N^3 sites, and nothing prevents it from being larger than r_q somewhere. The free field has typical size $q^{1/4} \ll r_q$, but it exceeds r_q with small, nonzero probability, and nothing a priori controls the interacting forecast. We need a force $\bar{\mathcal{F}}_q$ defined and *uniformly coercive for all fields*, which agrees with $\mathcal{F}_q^{\mathrm{o}}$ for $\|y\|_\infty \leqslant r_q$.

A scalar model. The difficulty is already visible in the one-site seed, $\ell_q(y) = \gamma y[(q+\gamma y^2)^{-1} - q^{-1}]$. Its power series in y has radius of convergence exactly r_q: the singularities are at $y = \pm i r_q$. Truncating it at any order produces a polynomial whose leading coefficient alternates in sign: half of the truncations destroy the monotonicity of $U' + \ell$ at $|y| \gg r_q$. The exact function is harmless: it is linear at large fields, and $|\ell_q'(y)| \leqslant \frac{9}{8}\gamma/q \leqslant 2\gamma/q$ for all y. In field theory we do not have the exact function, only bounds on the ball, obtained through an expansion of the loop in vertices (the corrections are not polynomials, but their estimates are expansions in the field). This is the large-field problem: perturbative control holds for $|y| \lesssim r_q$, and outside one needs a non-perturbative replacement that preserves the structure the estimates use.

The structure to preserve is coercivity, and it is carried by the bare cube. In the scalar model the coordinate equation $q\,y + \lambda\, y^3 = q\,\Phi_q$ has noise coefficient

$$H_q(y) = \frac{1}{q+\gamma y^2} \leqslant \frac{1}{q} \wedge \frac{1}{\gamma y^2};$$

large fields are *more* elliptic and less noisy. A good large-field extension keeps U untouched and freezes only the correction.

12.2 Clipping

Let $\tau : \mathbb{R} \to \mathbb{R}$ be smooth and odd, $\tau(u) = u$ on $[-1,1]$, constant outside $[-2,2]$, with $0 \leqslant \tau' \leqslant 1$, and set

$$\kappa_q(y)(x) := r_q\,\tau(y(x)/r_q), \quad \tilde\ell_q(y) := D\kappa_q(y)\,\ell_q^{[2]}(\kappa_q(y)), \quad \bar{\mathcal{F}}_q(y) := U'(y) + M_q y + \tilde\ell_q(y). \tag{12.1}$$

The clip acts pointwise in space and maps every field into the ball, and the factor $D\kappa_q$ is what keeps the clipped force a gradient. The cubic and linear parts are not clipped. By the chain rule, with factors $|\kappa_q^{(b)}| \leqslant C\, r_q^{1-b}$, the clipped correction obeys the same bounds (11.2) globally:

$$s(D^{j-1}\tilde\ell_q) \leqslant C\, g_q q\, r_q^{2-j} \quad (j = 2,3,4), \qquad s(D\partial_q \tilde\ell_q) \leqslant C\, g_q, \tag{12.2}$$

for all $y \in H_N$; the scale derivative includes the motion of the clip, $\partial_q \kappa_q = O(r_q/q)$.

Lemma 12.1. (Global coercivity) *For q_* large, for all $y \in H_N$ and $q \geqslant q_*$,*

$$\begin{aligned} c\ (K_q + \gamma y^2) \leqslant H_q(y)^{-1} &\leqslant C\ (K_q + \gamma y^2), \\ \|H_q(y)\|_{p\to p} \leqslant \frac{1 + C\,(g_q + g_q^2)}{q} \quad (1 \leqslant p \leqslant \infty), &\qquad \operatorname{diag} H_q(y)^2 \leqslant C\, a_2(q). \end{aligned} \tag{12.3}$$

Idea. $H^{-1} = K_q + \gamma y^2 + M_q + D\tilde\ell_q$, and $M_q + D\tilde\ell_q$ is small relative to K_q: $\|D\tilde\ell_q\| \leqslant C\, g_q\, q$, and M_q is small after multiplication by C_q (the centred kernel is controlled by its first moment, Exercise 12.1). Write $H = (I + G\,(M + D\tilde\ell))^{-1} G$ with $G = (K_q + \gamma y^2)^{-1}$ and use the Neumann series and Lemma 10.1. □

So $\bar{\mathcal{F}}_q$ is a globally coercive force, and Theorem 10.6 applies.

12.3 The complete defect

On the set $\{\|y\|_\infty \leqslant r_q\}$ the clip is the identity and the defect is $\mathcal{D}^{\mathrm{o}}$, of size (11.12). Outside it, the clipped force does not solve the force equation even approximately in relative terms. For $q \geqslant q_*$ split the drift into its bounded-field and large-field parts,

$$b_q^{\mathrm{b}}(y) := -H_q(y)\ D\kappa_q(y)\ \mathcal{D}_q^{\mathrm{o}}(\kappa_q y), \qquad b_q^{\mathrm{l}} := b_q - b_q^{\mathrm{b}},$$

and let $\chi_q(y) = 1_{|y|>r_q}$ (pointwise) and $t_q(y) = y\,\chi_q(y)$, the field restricted to the large-field set.

Proposition 12.2. (Complete source bounds) *For $q \geqslant q_*$, uniformly in $y \in H_N$, N and $p \in [1,\infty]$,*

$$\begin{aligned} &\|b_q^{\mathrm{b}}(y)\|_p \leqslant C\gamma^{7/2} q^{-5/2}, \\ &\|b_q^{1}(y)\|_p \leqslant C\gamma q^{-3/2}(\|t_q\|_p + r_q^{-1}\|y^2\chi_q\|_p), \\ &\|b_q^{1}(y)\|_2^2 \leqslant C\gamma^4 q^{-5}\|t_q\|_6^6. \end{aligned} \tag{12.4}$$

Moreover $\|b_q(y)\|_p \leqslant C\gamma q^{-3/2}\|y\|_p$.

The complete proof is Lemma 21.2 of the Supplement. The first bound is (11.13) with $n=2$: on the ball $|\mathcal{D}^{\mathrm{o}}| \leqslant \mathsf{S}_4(\mathcal{D}^{\mathrm{o}})\,(2r)^3/6$, and $q^{-1}\gamma^5 q^{-3} r_q^3 = \gamma^{7/2} q^{-5/2}$. The large-field part is as large as the bound $\gamma q^{-3/2}|y| = (g_q/q)|y|$ on the whole drift (no smallness beyond the factor g_q already present in b), but it is *supported on the large-field set* (up to nonlocal tails controlled by operator bounds). The last bound follows from the second since on $\{|y|>r\}$, $|t| + r^{-1}y^2 \leqslant 2r^{-2}|t|^3$, and $(\gamma q^{-3/2} r^{-2})^2 = \gamma^4 q^{-5}$.

Where the large-field error comes from. Compute the defect of the clipped force and compare it with $D\kappa\,\mathcal{D}^{\mathrm{o}}(\kappa y)$. With $z = \kappa_q(y)$, $J = D\kappa_q(y)$ (multiplication by $\tau'(y/r)$), $\ell = \ell_q^{[2]}$, $M' = \partial_q M_q$, and the unclipped response $H^{\mathrm{o}} = (K + M + \gamma z^2 + D\ell(z))^{-1}$, the difference has four groups (seven terms):

i. the mass term $M'y - JM'z$: the linear part is not clipped, but its flow was computed at z;

ii. the motion of the clip, $JD\ell(z)\partial_q\kappa + (D\partial_q\kappa)^*\ell(z)$;

iii. $\gamma(y - Jz)\mathrm{diag}H^2$ and $\gamma Jz\,\mathrm{diag}(H^2 - (H^o)^2)$: the cubic vertex sees y, the correction sees z, and the two responses differ by $\gamma(y^2 - z^2)$ and by the chain-rule terms of the clipped derivative, all supported on $|y| > r_q$;

iv. the contraction of $D^2\tilde{\ell} - JD^2\ell(z)$ (the chain rule produces terms with κ'', κ''', supported on large fields) and of $D^2\ell$ with $H^2 - (H^o)^2$.

Each group vanishes if $\|y\|_\infty \leqslant r_q$, and each is bounded by g_q times $|t|$ or $r^{-1}y^2\chi$, in L^p, by the same loop estimates (Section 21.3 of the Supplement; Exercise 12.2 for the one-site version). There is no cancellation to exploit: the clipping error is genuinely of the size $(g_q/q)|y|$ of the whole drift on the large-field set. What saves us is that this set is rarely visited, and its occupation is controlled by the coercivity of the actual dynamics. Lecture 6 controls it through the large-field action (13.7).

The coordinate correction. Finally, write the approximate coordinate equation as

$$K_q\Phi_q = K_q(Y_q + \varrho_q(Y_q)) + \tfrac{\gamma}{3}Y_q^3, \qquad \varrho_q(y) := C_q(M_q y + \tilde{\ell}_q(y)), \tag{12.5}$$

using $DU(y) = \lambda y^3 = \frac{\gamma}{3}y^3$. The correction is a small contraction, $\|D\varrho_q(y)\|_{p\to p} \leqslant \varepsilon_q := C(g_q + g_q^2)$, for all y and $1 \leqslant p \leqslant \infty$, and $\varrho_q(0) = 0$ by oddness. So, up to a small Lipschitz perturbation, *the forecast is the solution of the cubic elliptic equation $K_q(Y + \varrho) + \frac{\gamma}{3}Y^3 = K_q\Phi_q$ driven by the field at resolution q*. This is the equation whose coercivity Lecture 6 exploits.

12.4 Why the Polchinski expansion is not enough

Lecture 4 gives two exact descriptions of the flow: the Burgers equation for the force F at the field Φ, and the force equation at the forecast ψ. We use the second, here and in Lecture 6. The first looks more economical, and the following programme is natural. Keep the bare cube $-\lambda\varphi^3$ unclipped, so that the approximate force stays coercive at all fields. Expand the rest of the force in perturbation theory and clip only these higher orders. Pay for the clipping in the source of the equation for the remainder $R = F - f$, using the a priori estimates. No change of coordinates would be needed.

Here we test the programme where everything can be computed: on one site. There is no spatial structure and no renormalisation, so whatever goes wrong has nothing to do with divergences.

Conventions. As in Exercise 10.4: $N = 1$, $\Delta = 0$, $K_q = q$, $V(\varphi) = \frac{\lambda}{4}\varphi^4$, $\gamma = 3\lambda$ and $r_q = (q/\gamma)^{1/2}$. We write $S = C_q = 1/q$ for the remaining covariance, and x for the value of the field at resolution q.

The exact force is linear at large fields. The exact force is the conditional mean of the bare force,

$$F_q(x) = -\lambda \mathbb{E}_x Y^3,$$

where $\mathbb{E}_x$ is the expectation for the probability density proportional to $\mathrm{e}^{-\frac{\lambda}{4}y^4 - \frac{q}{2}(y-x)^2}$. It solves the Burgers equation

$$\partial_q F + \frac{1}{2}S^2 F'' + S^2 F F' = 0. \tag{12.6}$$

Let $m_q(x) = \mathbb{E}_x Y$. Integrating by parts in y gives

$$F_q(x) = -\lambda \mathbb{E}_x Y^3 = q(m_q(x) - x). \tag{12.7}$$

This is Proposition 10.4(b): the posterior mean is the forecast, $\psi_q = m_q(\Phi_q)$. So F_q measures how far the forecast lies behind the field.

Lemma 12.3. (Posterior mean) *For all x and q, $|m_q(x)| \leqslant 2\lambda^{-1/4} + 2(q|x|/\lambda)^{1/3}$.*

Proof. Integrating the derivative of y times the density gives $\lambda \mathbb{E}_x Y^4 + q\mathbb{E}_x Y^2 - q\, x\mathbb{E}_x Y = 1$. With $M = (\mathbb{E}_x Y^4)^{1/4}$ this gives $\lambda M^4 \leqslant 1 + q|x|M$, and Young's inequality gives $\frac{\lambda}{2}M^4 \leqslant 1 + (q|x|)^{4/3}(2\lambda)^{-1/3}$. Finally $|m_q| \leqslant M$. □

Therefore, by (12.7), the two regimes of the force are

$$F_q(x) \to -\lambda x^3 \quad (\text{fixed } x, q \to \infty), \qquad F_q(x) = -q\,x + O(|x|^{1/3}) \quad (\text{fixed } q, |x| \to \infty). \tag{12.8}$$

The limits cannot be interchanged. The crossover is at $|x| \sim r_q$, where the cube equals the Gaussian restoring force: $\lambda r_q^3 \sim q\, r_q$. Beyond it the force is linear, because convolution with a Gaussian of variance S cannot produce a force steeper than x/S.

Proposition 12.4. (The remainder of a frozen cube) *Let $f(x) = -\lambda x^3 + f_1(x)$ with $f_1(x) = o(|x|^3)$, for instance linear terms plus bounded clipped corrections. At every fixed q, the remainder $R_q = F_q - f$ satisfies*

$$\frac{R_q(x)}{\lambda x^3} \to 1 \quad (|x| \to \infty).$$

So no estimate can make R a small fraction of the retained cube uniformly in the field. The cube in R is not a new counterterm, since there is nothing to renormalise here. It is the compensation for having frozen the leading term of a force that does not stay cubic.

The field equation does not see the cube. The field equation (18.20) of Lecture 6 on one site is $q_0\Phi_q = J_q F_q(\Phi_q) + \eta_q$, where $J_q = (q - q_0)/q$. Its noise is the conditional expectation of the bridge noise, so $\mathbb{E}\exp(h\eta_q) \leqslant \exp((q-q_0)h^2/2)$ whatever the interaction (Lemma 13.2 of Lecture 6). With the frozen cube $f = -\lambda x^3$ the equation looks coercive:

$$q_0 x + J_q \lambda x^3 = J_q R_q(x) + \eta_q(x).$$

But by Proposition 12.4, $J_q R_q$ cancels the cube on the left side at large fields. Inserting (12.7) shows what is left:

$$q\,x = (q - q_0) m_q(x) + \eta_q(x). \tag{12.9}$$

There is still confinement, but it is linear, and it comes from the sublinear growth of m_q.

Proposition 12.5. (Tails on one site) *If $q^2 \geqslant 24\lambda$, then $|m_q(x)| \leqslant \frac{9}{10}|x|$ for $|x| \geqslant r_q$, and for $R \geqslant r_q$*

$$\mathbb{P}(|\Phi_q| \geqslant R) \leqslant 2\exp\Big(-\frac{q^2 R^2}{200(q-q_0)}\Big).$$

Proof. The mean m has the sign of x, the density being an even density tilted by e^{qxy}. Jensen's inequality in the identity of Lemma 12.3 gives $\lambda m^4 + q m^2 - q\,x\,m \leqslant 1$. If $x \geqslant r_q$ and $m > \frac{9}{10}x$, then $\lambda m^3 + q(m-x) > x\big(\frac{729}{1000}\lambda x^2 - \frac{q}{10}\big) \geqslant \frac{143}{1000} q\,x$, as $\lambda x^2 \geqslant q/3$, so the left side is at least $\frac{1287}{10000} q\,x^2 \geqslant \frac{429}{10000} q^2/\lambda > 1$, a contradiction. So $|\eta_q(x)| \geqslant \frac{q}{10}|x|$ by (12.9), and we conclude with the sub-Gaussian bound. □

At $R = r_q$ the exponent is at least $q^2/(600\lambda)$. This estimate treats the combined restoring force through the posterior moment. It never uses the cube as a coercive term against a remainder regarded as a source.

Remark 12.6. (This does not transfer to the lattice) With a linear term $a\,y$ in the force, Lemma 12.3 acquires the term $((q+a)_-/\lambda)^{1/2}$. On the lattice the bare mass makes a divergent and negative, so this elementary moment bound is not uniform until the renormalised cancellations have been built in. The scalar tail bound is therefore not an independent input for the field theory.

What clipping does to the Burgers equation. For a force f put $D_q = \partial_q + \frac{1}{2}S^2\partial_x^2$ and let $E_q(f) = D_q f + S^2 f f'$ be its Burgers defect, so $E_q(F_q) = 0$. Perturbation theory solves

$$D_q f_1 = 0, \qquad D_q f_2 = -S^2 f_1 f_1',$$

with $f_1 = -\lambda(x^3 + 3S\,x)$, the cube and its Wick correction, and $f_2 = 3\lambda^2 S\,x^5 + \ldots$, the first tree. Keep f_1 unclipped and clip the correction: $f^c = f_1 + \mathcal{C} f_2$ with $\mathcal{C} f_2 = \kappa_q' \cdot f_2 \circ \kappa_q$, the clip κ_q of (12.1). Then

$$E_q(f^c) = (1 - \mathcal{C}) S^2 f_1 f_1' + [D_q, \mathcal{C}] f_2 + S^2 (f_1 \mathcal{C} f_2)' + S^2 \mathcal{C} f_2 (\mathcal{C} f_2)'. \tag{12.10}$$

Without the clip only the last two terms, of higher order, would be present. The clip reintroduces a defect at exactly the order that f_2 was built to cancel. For $|x| \geqslant 2r_q$ we have $\mathcal{C} f_2 = 0$, so $f^c = f_1$, and the Burgers defect is exactly

$$E_q(x) = S^2 f_1(x) f_1'(x) = 3\lambda^2 S^2 x^5 (1 + O(S/x^2)). \tag{12.11}$$

This is the price of the clipping. At the edge of the clip it has size $(q/\lambda)^{1/2}$, which is not integrable in q, so it can only be paid through the small probability of reaching the clip. Along the exact process $\mathrm{d}\Phi_q = S^2 F_q(\Phi_q)\,\mathrm{d}q + S\,\mathrm{d}W_q$, Itô's formula gives for $R_q = F_q(\Phi_q) - f_q^c(\Phi_q)$ the equation

$$\mathrm{d}R_q = -[E_q(\Phi_q) + S^2 (f_q^c)'(\Phi_q) R_q]\,\mathrm{d}q + \Xi_q^R\,\mathrm{d}W_q. \tag{12.12}$$

The transport term has a favourable sign at large fields: $(f^c)' = -3\lambda(x^2+S)$ for $|x| \geqslant 2r_q$, which is the dissipation one hopes to use (near the clip the sign fails: on the ball the quintic takes over beyond $|x| \approx 0.77 r_q$). But at large fields the bracket is $3\lambda^2 S^2 x^5 - 3\lambda S^2 x^2 R$, and for the true $R \sim \lambda x^3$ the two terms cancel at leading order. The source and the dissipation are the same quantity, so estimating them separately gives nothing. To close, one would need, for instance, a signed estimate that retains this cancellation, or a tail bound for Φ obtained independently of R. On one site Proposition 12.5 is such a bound: the probability of reaching the clip is at most of order $\exp(-c\,q^2/\lambda)$, so in expectation the source is exponentially small, and the frozen cube is not ruled out there. But that bound comes from the posterior moment, and by the remark above it does not transfer. The field equation alone cannot replace it, since its restoring force is the cube and R together, which is what we are trying to estimate.

Going further in perturbation theory does not help: any finite depth of expansion leaves the same structure (Exercise 12.5), and we shall see in a moment that the series of trees stops converging inside r_q.

Integrating the cube. The term responsible for all this is the nonlinear transport in (12.6). So solve it exactly: let f_q solve the Burgers equation without diffusion,

$$\partial_q f + S^2 f f' = 0, \qquad f_\infty(x) = -\lambda x^3. \tag{12.13}$$

The characteristics give the solution in closed form.

Proposition 12.7. (The integrated cube) *For $x \in \mathbb{R}$, let $y = y_q(x)$ be the unique real root of $q\,y + \lambda y^3 = q\,x$. Then the solution of (12.13) is*

$$f_q(x) = -\lambda\, y_q(x)^3 = q(y_q(x) - x), \tag{12.14}$$

and it has the following properties.

- *a)* ***Variational form.*** *$y_q(x)$ is the minimiser of $\frac{\lambda}{4}y^4 + \frac{q}{2}(y-x)^2$, and $f_q = -\bar{V}_q'$ for the minimum value $\bar{V}_q(x)$.*
- *b)* ***Small fields.*** *$f_q(x) = q\sum_{k\geqslant 1} (-\lambda/q)^k c_k x^{2k+1}$ with $c_k = \frac{1}{2k+1}\frac{(3k)!}{k!(2k)!}$, so $f_q = -\lambda x^3 + 3\lambda^2 S\, x^5 - 12\lambda^3 S^2 x^7 + \dots$. This is the sum of the tree expansion. Its radius of convergence is $|x| = \frac{2}{3} r_q$.*
- *c)* ***Large fields.*** *$\lambda|y|^3 \leqslant q|x|$, so $f_q(x) = -q\,x + O(|x|^{1/3})$ at fixed q, as for the exact force, and $f_q' = -\gamma y^2/(1+\gamma y^2/q) \leqslant 0$ for all x.*
- *d)* ***Defect.*** *f_q fails to solve Burgers by the diffusion term alone, which is q times the drift of Exercise 10.4:*

$$E_q = \frac{1}{2} S^2 f_q''(x) = q\, b_q(y_q(x)), \qquad b_q(y) = -\gamma y (q + \gamma y^2)^{-3}. \tag{12.15}$$

Proof. Differentiate $x = y + \lambda S\, y^3$ with $d = 1 + \gamma y^2/q$: $y' = 1/d$, $\partial_S y = -\lambda y^3/d$, $f' = -\gamma y^2/d$ and $f'' = -2\gamma y/d^3$. Hence $\partial_S f = \gamma\lambda y^5/d = f f'$, which is (12.13) as $\partial_q = -S^2\partial_S$. For (a), the stationarity condition is $\lambda y^3 + q(y-x) = 0$. For (b), use Lagrange inversion on $y = x - \lambda S y^3$; the nearest singularity is where $d = 0$, that is at $y = \pm \mathrm{i} r_q$, $x = \pm\frac{2}{3}\mathrm{i} r_q$. Finally (c) and (d) are read off from the formulas, since $d = 1/(q\,H_q(y))$ with $H_q(y) = (q+\gamma y^2)^{-1}$. □

Item (b) explains why the depth of the expansion is not the issue. The tree series has the same complex singularity $y = \pm i r_q$ as the one-loop seed of Section 12.1, and its truncations at even k point outward at large fields.

The defect is now bounded for all fields, and it gives a global estimate.

Theorem 12.8. (Global error of the integrated cube) *Let* $C_* = 3\sup_z |z|(1+3z^2)^{-3}$, *so* $C_* < \frac{1}{2}$. *For all* $q > 0$,

$$\sup_{x\in\mathbb{R}} |F_q(x) - f_q(x)| \leqslant 2C_*(\lambda/q)^{1/2}. \tag{12.16}$$

Proof. Use $S = 1/q$ as time, starting from $S = 0$. Subtracting, $\hat{R} = F - f$ solves the linear problem

$$\mathfrak{L}\hat{R} = \frac{1}{2}f'', \quad \mathfrak{L} = \partial_S - \frac{1}{2}\partial_x^2 - F\partial_x - f', \quad \hat{R}_0 = 0.$$

The zeroth-order coefficient has the good sign, as $-f' \geqslant 0$. With $z = (\lambda S)^{1/2} y$, the source obeys $\frac{1}{2}|f''| \leqslant C_*(\lambda/S)^{1/2}$, which is integrable at $S = 0$, so $\beta(S) = 2C_*(\lambda S)^{1/2}$ satisfies $\mathfrak{L}\beta \geqslant \frac{1}{2}|f''|$. To apply the maximum principle on the line, take $\chi = e^{3S}(1+x^4)$. Then $\mathfrak{L}\chi \geqslant 0$, since $6x^2 \leqslant 3(1+x^4)$ and $xF(x) \leqslant 0$; the last holds because $F = -\lambda \mathbb{E}_x Y^3$, and $\mathbb{E}_x Y^3$ has the sign of x, the density being an even density tilted by e^{qxy}. Both F and f grow at most cubically, uniformly for bounded S: $|f| \leqslant \lambda|x|^3$ since $|y| \leqslant |x|$, and $\lambda \mathbb{E}_x Y^6 + q\mathbb{E}_x Y^4 - q\,x\mathbb{E}_x Y^3 = 3\mathbb{E}_x Y^2$ gives $\mathbb{E}_x Y^4 \leqslant (|x| + (3S)^{1/2})^4$, up to constants, hence $|F| \leqslant C\lambda(|x|^3 + S^{3/2})$. Also $\hat{R}$ is continuous at $S = 0$, since the posterior concentrates at x. So $\pm\hat{R} - \beta - \varepsilon\chi$ is negative on the boundary of a large cylinder, hence inside it. Let the cylinder grow and then $\varepsilon \to 0$. □

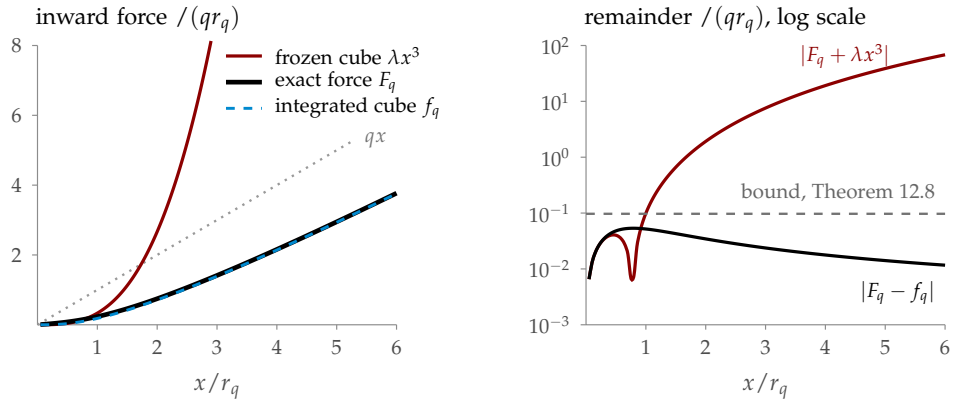


Figure 12.1. The force on one site, for $\lambda = 1$, $q = 4$. Left: the frozen cube leaves the exact force at $x \sim r_q$, while the integrated cube follows it at all fields: the two curves coincide at this scale, their difference being at most 0.06 in these units. Right, on a logarithmic scale: the remainder of the frozen cube grows like the cube itself, while the remainder of the integrated cube stays below the bound of Theorem 12.8.

The integrated cube is the forecast coordinate. The equation $qy + \lambda y^3 = qx$, at $x = \Phi_q$, is the coordinate equation of Exercise 10.4, so $y_q(\Phi_q) = Y_q$. Equations (12.7) and (12.14) then say that $F_q - f_q = q(m_q - y_q)$: the remainder is $K_q Z_q$, and the error Z of (10.11) is the posterior mean minus the posterior mode. By (12.16), on one site $|Z_q| \leqslant 2C_*\lambda^{1/2}q^{-3/2}$ surely, for all fields.

Why the forecast coordinate has no such problem. In the variable y the bare cube does not flow at all: all the trees of (b) are contained in the map $x \mapsto y$. The second term of (10.9) is a loop, and the defect $\mathcal{D}_q(y) = \gamma y H_q(y)^2$ of $\bar{\mathcal{F}}_q = \lambda y^3$ is bounded by $\gamma^{1/2}q^{-3/2}$ for all y. This is why Section 12.2 can leave the cube untouched and clip only the corrections: what is clipped consists of loops, not of trees whose large-field sum is forced to change the growth of the force. This removes the frozen-cube problem specifically: the clipping error of Section 12.3 is still there, and it is what Lecture 6 pays for.

In the coordinate y the cube is genuine: $q\,y+\lambda y^3 = q\,\Phi_q = (q-q_0)\psi_q + \eta_q$, i.e. $q_0 y + \lambda y^3 = \eta_q + (q-q_0)(\psi_q - y)$. This is the forecast equation of Lecture 4 with the explicit force in place of the exact one, the equation (13.3) of Lecture 6 on one site.

What the model points to. The scalar model points to the integration of the cube. The force equation in the forecast coordinate and the FBSDE of Theorem 10.6 are one way to organise it; another would keep the original field, with the force f_q given by the variational formula of (a), and work with the defect $\frac{1}{2}\mathrm{Ctr}_{C_q^2} D^2 f_q$. On the lattice the variational formula still defines this force, with a nonlocal response, but its defect contains the loop contractions, starting with the tadpole, and must be renormalised. A complete proof along that route is not given in these lectures.

The scalar model proves less than a no-go theorem. It shows that a global estimate treating the remainder of a frozen cube as a small cubic perturbation is false. It does not show that every construction based on the Polchinski equation fails, nor does it give the asymptotics of the force on the lattice. There is also a limit to the integrated cube itself. With a linear term in the force, the map $y \mapsto y + S(\lambda y^3 + r y)$ has derivative $1 + S(r + 3\lambda y^2)$, which vanishes somewhere when $1 + S\,r < 0$ (Exercise 12.6). A divergent negative mass cannot simply be placed inside this monotone map, and this is one reason why the mass correction M_q of (12.1) is kept apart from the cube and is required to be small relative to K_q (Lemma 12.1).

12.5 The force below the auxiliary scale

The force above is built for $q \geqslant q_*$, because its expansion needs g_q small. Nothing below q_* requires a construction: we freeze it. For $0 < q < q_*$ set

$$\bar{\mathcal{F}}_q(y) = \bar{\mathcal{F}}_{q_*}(y) + (q_* - q)y. \tag{12.17}$$

Then $K_q + D\bar{\mathcal{F}}_q = K_{q_*} + D\bar{\mathcal{F}}_{q_*}$, so the coordinate equation $K_q y + \bar{\mathcal{F}}_q(y) = K_q x$, at $x = \Phi_q$, reads $K_{q_*} y + \bar{\mathcal{F}}_{q_*}(y) = K_q x$. Thus for $q < q_*$,

$$H_q = H_{q_*}, \quad y_q(x) = y_{q_*}(C_{q_*} K_q x), \quad C_{q_*} K_q = I - (q_* - q) C_{q_*}. \tag{12.18}$$

The Itô computation of Theorem 10.6 holds for any such family. So the system (10.12) holds on $[q_0, \infty)$, with $Y_{q_0} = 0$ for ν_N: the initial value is deterministic, and no large mass is used. The force is continuous in q and piecewise C^1, which is all that Itô's formula needs. The defect for $s < q_*$ is explicit:

$$\mathcal{D}_s(y) = -y + \mathcal{U}(y), \quad \mathcal{U}(y) = \frac{1}{2}\mathrm{Ctr}_{H_{q_*}(y)^2} D^2 \bar{\mathcal{F}}_{q_*}(y). \tag{12.19}$$

Here $\mathcal{U}$ is the loop part of $\mathcal{D}_{q_*}$; it obeys $\|\mathcal{U}(y)\|_p \leqslant C\, g_{q_*} \|y\|_p$, which is part of the proof of the global linear bound of Proposition 12.2 (Lemma 21.2 of the Supplement). Uniformly in N, for $s < q_*$,

$$\|\mathcal{D}_s(y)\|_p \leqslant C\|y\|_p, \quad \|b_s(y)\|_p \leqslant C\|y\|_p, \quad \mathrm{diag} H_s^2 \leqslant C, \tag{12.20}$$

with $b = -H\mathcal{D} = H(y - \mathcal{U})$. Here we used $\|H\|_{p\to p} \leqslant C/q_*$. These are all the properties of $s < q_*$ we use. Note that $\mathcal{D}$ jumps at q_*; the jump is harmless, and the measure does not depend on q_*. More precisely $\bar{\mathcal{F}}, y, \mathcal{D}, b, Y, Z$ depend on q_*, while, with μ_N fixed (b_N then depends on q_*, Theorem 11.2), ν, ψ, Cov and the law of Φ do not.

Exercises

Exercise 12.1. (Receiving a centred kernel) Let Θ be a translation-invariant, even kernel with $\Theta 1 = 0$. With T_z the translation by z, $(T_z f)(x) = f(x+z)$, and using $C_q \Theta = \frac{1}{2}\int \Theta(z)\,(T_z + T_{-z} - 2)\,C_q\,\mathrm{d}z$, show that $\|C_q\Theta\|_{\mathrm{row}} \leqslant C\, q^{-1/2} v_1(\Theta)$ if $\|(T_z + T_{-z} - 2)\,C_q\|_{\mathrm{row}} \leqslant C q^{-1/2} d(0,z)$. Why is the absolute row mass of Θ not the right quantity?

Exercise 12.2. (**The scalar clipping error**) On one site, let $\ell(y) = -\beta\gamma^2 q^{-2} y^3$ (a cubic correction, β a fixed number) and $\bar{\mathcal{F}}_q = \lambda y^3 + \kappa_q{}'(y)\,\ell(\kappa_q(y))$. (a) Show that $K_q + \bar{\mathcal{F}}_q{}' \geqslant q/2$ for all y if $\beta\gamma \leqslant c\,q^2$. (b) Show that $b_q = b_q^b$ for $|y| \leqslant r_q$. For $|y| > 2r_q$ show that κ_q is locally constant, $b_q^b(y) = 0$, and $b_q(y) = -\gamma y (q + \gamma y^2)^{-3}$, the drift of the pure cube. (c) Conclude that on the large-field set the error is the whole drift of the loop term: it is comparable, for $|y| \approx 2r_q$, to the one-site bound $\gamma q^{-3}|y|$ on the whole drift (the analogue of $(g_q/q)|y|$, with $a_2(q) = q^{-2}$ in place of $q^{-1/2}$), but it vanishes on the ball. Compare with (12.4).

Exercise 12.3. (**Large fields are less noisy**) On one site with $\bar{\mathcal{F}} = \lambda y^3$, show that the quadratic variation of Y is $\int H_q(Y_q)^2\,\mathrm{d}q$ and that $H_q(y)^2 \leqslant (\gamma y^2)^{-2}$. Simulate the Polchinski process for $\nu = \mathrm{e}^{-\lambda\varphi^4/4}\mu^{1/q_0}$ and plot Φ_q, Y_q and ψ_q.

Exercise 12.4. (**The defect**) Check (12.15) and integrate it: show that $\int_q^\infty q'\sup|b_{q'}|\,\mathrm{d}q'$ equals the right side of (12.16). Why does the integration of the cube turn the x^5 of (12.11) into a bounded function?

Exercise 12.5. (**Depth of the expansion**) Let f be any approximate force, $f^c = f + \delta$. Show that $E_q(f^c) = E_q(f) + D_q\delta + S^2(f\delta)' + S^2\delta\delta'$. Which terms are controlled by the smallness of the truncation defect $E_q(f)$? Take f the tree expansion up to order k, and δ the clipping of its terms beyond the cube. What replaces $3\lambda^2 S^2 x^5$ in (12.11)?

Exercise 12.6. (**Negative mass**) For $r < 0$, let $y \mapsto y + S(\lambda y^3 + r y)$. Show that it is monotone if and only if $1 + Sr \geqslant 0$. What happens to the minimiser in the variational formula when $1 + Sr < 0$, and to the solution of (12.13) with $f_\infty = -\lambda x^3 - r x$? Why does this not contradict the existence of the exact, smooth force F_q?

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